# Leo Li

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## Education

**University of Waterloo** Sep 2026 – Apr 2031
*Bachelor of Applied Science in Computer Engineering* *Waterloo, ON*
**Relevant Coursework:** *Fundamentals of Programming (C++), Linear Algebra, Calculus I*

## Skills

**Languages**: C++, Python, Java, TypeScript, JavaScript, SQL, C#, HTML/CSS
**Frameworks**: React, Node.js, PyTorch, Jupyter
**Platforms**: Git, Unity

## Projects

**BetaGo** | *C++20, MCTS, CNN, Tk*

- Built a **9x9 Go engine** with capture, suicide, ko, and area scoring rules; added an interactive Tk board and **seeded classical MCTS** search
- Developed a **CPU policy/value CNN** to guide neural MCTS and a **resumable self-play pipeline** trained on search visits and game outcomes
- Optimized liberty checks and board copying, reducing five 64-simulation searches from 312 ms to 115 ms (**2.70x**) and 32-move self-play from 1.75 s to 0.91 s (**1.94x**)

**Video to ASCII** | *TypeScript, React, WebGL2, GLSL*

- Built a reusable **video-to-ASCII component** with a two-pass **WebGL2 pipeline**, using six-region glyph matching to render grids of up to **500 columns**
- Capped real-time playback at **30 FPS** and reduced GPU overhead through texture reuse, offscreen suspension, and reliable WebGL context recovery
- Implemented **browser-local video export** at up to **1920 px** on the longest side; validated playback and export with **40 passing unit tests** and Edge browser tests

**Did You Even Listen** | *TypeScript, Obsidian API, Groq, esbuild*

- Built an **Obsidian plugin** that records lectures, compares student notes with AI-generated notes, and provides targeted **Socratic feedback** to improve learning
- Implemented **real-time lecture transcription** using **Groq Whisper** and AI-powered note analysis, metacognition assessment, and quiz generation using **Llama 3.3**
- Designed **structured JSON prompts** and feedback workflows that identify missing, incorrect, vague, and verbose notes without directly revealing answers

## Experience

**Undergraduate Researcher** Nov 2026 – Present
*University of Waterloo — Fast Multipole Attention* *Waterloo, ON*

- Investigating **Fast Multipole Attention (FMA)** under **Professor Giang Tran**, exploring efficient Transformer architectures to reduce the computational complexity of standard self-attention
- Implementing **FlashAttention** using **blockwise computation and online softmax** to reduce GPU memory overhead by avoiding materialization of the full attention matrix

**Software Engineer** Sep 2026 – Present
*UW Reality Labs* *Waterloo, ON*

- Built a **Python validation tool** for gesture-recognition datasets, auditing **372 recordings across 22 JSON files** and reporting data-quality issues with precise source locations
- Implemented **5 dataset validation checks**, covering missing sequence and confidence fields, empty recording lists, invalid record types, and empty recording objects
- Implemented **sliding-window segmentation, confidence filtering, and sequence padding** to prepare gesture data for ML training.